

# **GE107: TINKERING LAB**

## **PROJECT REPORT**



## **SMART DOOR PEEPHOLE CAMERA SYSTEM**

**PROJECT GROUP NO : 34**



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# OVERVIEW

The Smart Door Peephole Camera system is an innovative solution designed to enhance home security by modernizing the traditional door peephole. In an era where safety is a growing concern, this project aims to provide homeowners with a comprehensive surveillance tool that combines real-time monitoring, motion detection, and remote access capabilities. By integrating a camera within a conventional peephole, the system transforms the door into an advanced security checkpoint.

The core idea is to use a compact, efficient Raspberry Pi board as the brain of the system, paired with a camera module to capture high-quality video footage of visitors. A motion detection sensor acts as the system's trigger, activating the camera when movement is detected. This ensures that the system is energy-efficient and only operational when necessary. The addition of internet connectivity further enhances the system's functionality by allowing users to remotely access live video feeds, receive notifications, and store footage for future reference.

Unlike traditional peepholes, which require the homeowner to be physically

present to view the visitor, this smart system offers convenience and accessibility. With features like optional LCD displays for live feedback and secure data storage using a MicroSD card, the Smart Door Peephole Camera system represents a significant leap forward in residential security technology.

The project leverages advancements in Internet of Things (IoT) and embedded systems, aiming to deliver a cost-effective and user-friendly solution. Whether for urban apartments or standalone houses, this system provides homeowners with peace of mind, knowing they have a reliable tool to monitor and secure their entrance. This innovation is particularly valuable in addressing the limitations of conventional security measures, bridging the gap between traditional peepholes and high-end surveillance systems.

# MOTIVATION

The motivation behind developing the Smart Door Peephole Camera system stems from the growing need for enhanced home security in modern society. Traditional door peepholes, though widely used, have significant limitations. They require the homeowner to be physically present at the door to see who is outside, and they provide only a narrow, static field of view. Furthermore, they lack any means of recording or alerting the homeowner to suspicious activity, leaving security gaps that can be exploited.

The increasing rate of doorstep thefts, unauthorized access attempts, and the rise of smart homes have highlighted the need for an innovative solution. Many homeowners seek affordable and efficient ways to monitor their homes, especially when they are away. This need is even more pressing in urban areas, where individuals may feel more vulnerable due to higher crime rates. The Smart Door Peephole Camera system was conceptualized as a direct response to these concerns, combining accessibility, affordability, and technological sophistication. Additionally, the project is motivated by the potential to democratize smart security systems. While advanced surveillance solutions are available in the market, they often come with high costs and complex installation procedures. This project aims to offer a simple, do-it-yourself system that can be easily implemented by individuals with minimal technical expertise.

Another key inspiration is the integration of IoT technology into daily life. The idea of connecting a simple peephole to the internet to enable remote access and control was both a challenge and an opportunity to explore the practical applications of embedded systems and real-time data transfer. The Smart Door Peephole Camera system not only ensures safety but also empowers homeowners by giving them full control over their door security, whether they are at home or away.

The ultimate goal of the project is to combine safety, convenience, and technological advancement in one affordable package, ensuring that everyone has access to reliable home security.

# OBJECTIVE

The primary objective of the Smart Door Peephole Camera system is to enhance home security by modernizing the traditional peephole with smart technologies. This project focuses on addressing the limitations of conventional peepholes by providing an innovative solution that is both effective and user-friendly. The objectives of this project are as follows:

1. **Real-Time Monitoring:** To allow homeowners to see visitors at their door in real-time through live video streaming, offering better situational awareness and security.
2. **Motion Detection:** To integrate a motion detection sensor that triggers the system to start recording or alert the homeowner when movement is detected near the door.
3. **Remote Accessibility:** To establish internet connectivity, enabling homeowners to remotely access live feeds, receive alerts, and control the system via their smartphones or computers.
4. **Data Storage:** To provide local data storage using a MicroSD card for recording video footage, which can be reviewed later if needed.
5. **Cost-Effective Design:** To ensure the system is affordable and accessible to a wide range of users, making smart security available to everyone.
6. **Ease of Use:** To make the system intuitive and simple to install and operate, ensuring it can be used by individuals with minimal technical expertise. By achieving these objectives, the Smart Door Peephole Camera system aims to offer a reliable, modern, and efficient security solution for residential spaces, empowering homeowners with greater control over their safety.

# WORKING PRINCIPLE

The Smart Door Peephole Camera system operates through the seamless integration of hardware and software components to deliver enhanced security. Its working principle can be explained in the following steps:

## 1. **Motion Detection:**

- A passive infrared (PIR) motion detection sensor monitors the area outside the door. When it detects movement, it sends a signal to the Raspberry Pi to activate the system.

## 2. **Video Capture:**

- Upon activation, the Raspberry Pi powers the connected camera module. The camera captures live video footage of the area outside the door, ensuring a clear view of visitors.

## 3. **Data Processing and Storage:**

- The Raspberry Pi processes the video feed and stores it on a MicroSD card for local storage. This enables users to review recordings if necessary.

## 4. **Internet Connectivity:**

- The system uses Wi-Fi or Ethernet to establish an internet connection. This allows homeowners to access the video feed remotely through a smartphone or computer.

## 5. **Alerts and Notifications:**

- When motion is detected, the system can send notifications or alerts to the homeowner, ensuring they are informed of any activity at their door.

## 6. **Optional LCD Display:**

- An optional LCD display can be included to provide real-time feedback on the system's status or to show live video feeds directly at the door. This workflow ensures that the Smart Door Peephole Camera system is both efficient and reliable, providing continuous monitoring and recording capabilities while being energy-efficient and easy to operate.

# COMPONENTS USED:

The Smart Door Peephole Camera system utilizes the following components to achieve its functionality:

1. **Raspberry Pi Board:** Acts as the central processing unit, controlling the system's operations and managing data.
2. **Camera Module:** Captures live video footage. A compatible Raspberry Pi camera module or a USB webcam can be used.
3. **Motion Detection Sensor (PIR):** Detects movement outside the door and triggers the system to start recording.
4. **MicroSD Card:** Provides local storage for video recordings and system data.
5. **Internet Connectivity Module:** Wi-Fi or Ethernet connectivity enables remote access and control.
6. **LCD Display (Optional):** Offers live feedback on the system's status and displays video streams, enhancing usability.
7. **Power Supply:** Ensures a stable power source to keep the system operational at all times.

These components are carefully chosen to balance cost, performance, and ease of use, ensuring the system is accessible to a broad audience.

# SYSTEM WORKFLOW

The workflow of the Smart Door Peephole Camera system integrates hardware components and software functionality to ensure efficient operation. Below is a step-by-step description of how the system works:

## 1. Standby Mode

- The system remains in standby mode to conserve power and is activated only when the motion detection sensor (PIR sensor) detects movement near the door.

## 2. Motion Detection

- When motion is detected, the PIR sensor sends a signal to the Raspberry Pi. This triggers the system to initiate video capture.

## 3. Video Capture and Processing

- The Raspberry Pi activates the connected camera module to capture live video footage.

- The video feed is processed in real-time by the Raspberry Pi to ensure clarity and stability.

#### **4. Data Storage**

- The captured video footage is stored locally on a MicroSD card. This allows the homeowner to review the footage later if needed.

#### **5. Internet Connectivity for Remote Access**

- If the Raspberry Pi is connected to the internet via Wi-Fi or Ethernet, the system uploads the video feed to a remote server or allows live streaming through a mobile app or web interface.
- Homeowners can access the video feed remotely using their smartphones, computers, or other devices.

#### **6. Alerts and Notifications**

- Upon detecting motion, the system can send notifications to the homeowner via email, SMS, or push notifications (if configured).
- These alerts ensure the homeowner is immediately informed of activity at the door, even when they are not home.

#### **7. Optional LCD Display (Real-Time Monitoring)**

- If an LCD display is integrated into the system, it shows the live video feed directly at the door. This allows homeowners to view visitors without the need for remote devices.

#### **8. Return to Standby Mode**

- After the motion ceases and a preset time period elapse, the system stops recording and returns to standby mode, ready for the next activation.

## RESULTS

The Smart Door Peephole Camera system successfully achieves its objectives, providing a reliable and efficient home security solution. The results of the project are as follows:

1. **Reliable Video Monitoring:** The camera module captures clear video footage, offering a comprehensive view of visitors at the door.
2. **Effective Motion Detection:** The PIR sensor accurately detects movement, minimizing false triggers and ensuring timely alerts.
3. **Remote Access:** The system's internet connectivity allows homeowners to monitor their door remotely, adding convenience and flexibility.
4. **User-Friendly Design:** The intuitive interface and simple installation process make the system accessible to users with minimal technical expertise.
5. **Data Storage:** The MicroSD card provides ample space for storing video recordings, ensuring that users can review footage when needed.

The system's performance demonstrates its potential to enhance home security effectively while maintaining affordability and ease of use.

## CHALLENGES FACED

During the development of the Smart Door Peephole Camera system, several challenges were encountered:

1. **Hardware Compatibility:** Ensuring that all components, such as the camera module and motion sensor, were compatible with the Raspberry Pi board was a significant challenge.
2. **Motion Detection Accuracy:** Tuning the PIR sensor to reduce false alarms caused by non-human movements, such as animals or wind, required careful calibration.
3. **Internet Connectivity:** Maintaining a stable Wi-Fi or Ethernet connection was critical for remote access, and resolving connectivity issues was timeconsuming.
4. **Power Consumption:** Optimizing the system to minimize power usage while ensuring continuous operation posed a challenge.
5. **Video Quality and Storage Management:** Balancing video quality with storage space availability required careful consideration of recording settings and compression techniques.

Despite these challenges, the project successfully addressed these issues, resulting in a robust and functional smart peephole camera system.

## CONCLUSION

The Smart Door Peephole Camera system successfully enhances home security by transforming a traditional peephole into a modern, intelligent monitoring solution. With features like real-time video recording, motion detection, and remote access, it provides homeowners with an affordable and efficient way to ensure safety. Future upgrades can include advanced capabilities like face recognition and mobile app integration, making it even more versatile and user-friendly.